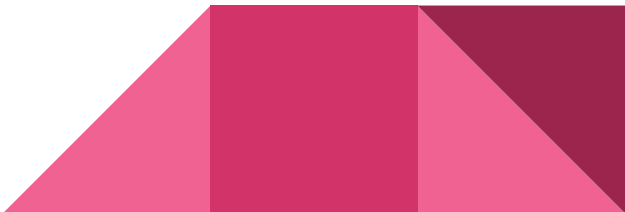


SoftNet Lab Cluster

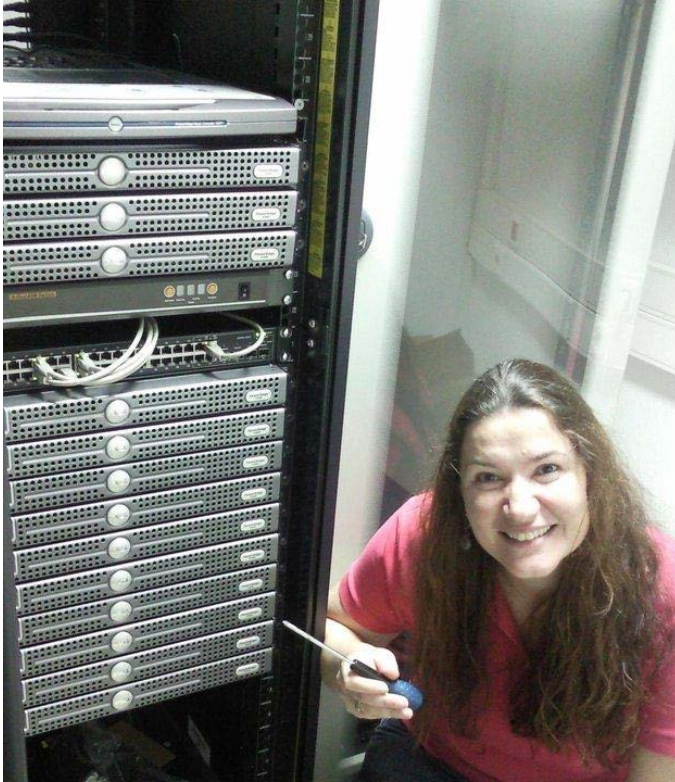
History, structure and management, access and job submission (Hadoop MapReduce, Spark, Flink), Kafka messages

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Lecture structure

- SoftNet cluster history and infrastructure
 - Hortonworks Data Platform (HDP)
 - HDP on SoftNet cluster managed by Ambari
 - YARN Resource Manager UI / Applications
 - Submitting/Running jobs on SoftNet cluster (Hadoop MapReduce, Spark, Flink)
 - Kafka (create / list / delete topics, producing / consuming messages)
 - Hortonworks Sandbox for HDP installation and use
 - Running Spark, Flink jobs
- 

SoftNet cluster - When/How it started



- 2009 - 5 x Dell PowerEdge R300
Quad Core Xeon X3323 2.5GHz, 4GB RAM, 500GB HDD
- 2010 - 8 x Dell PowerEdge R300
Quad Core Xeon X3323 2.5GHz, 4GB RAM, 500GB HDD

SoftNet cluster servers

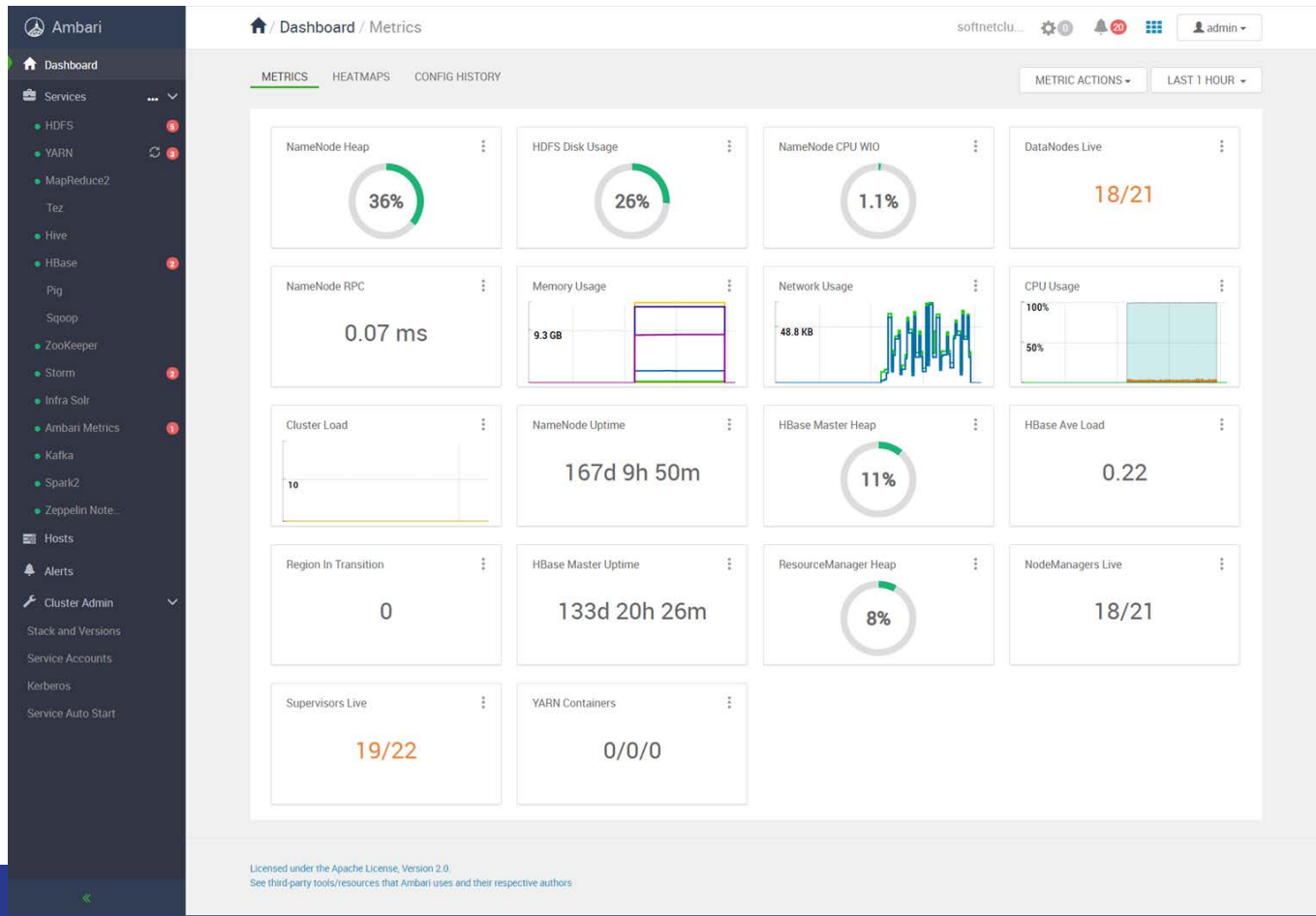
Nowadays, the SoftNet cluster is hosted in NOC Data Center and consists of **23 servers**:

- **13 x Dell PowerEdge R300 Quad Core Xeon X3323 2.5GHz, 8GB RAM, 500GB HDD** {clu11, clu12, clu13, **clu14**, clu15, clu16, clu17, **clu18**, clu19, clu20, clu24, clu25, clu26} - 2009/10
- **3 x Dell PowerEdge R310 Quad Core Xeon X3440 2.53GHz, 8GB RAM, 500GB HDD** {clu08, clu09, **clu10**} - 2011
- **3 x Dell PowerEdge R310 Quad Core Xeon X3440 2.53GHz, 16GB RAM, 500GB HDD** {clu05, clu06, **clu07**} - 2011
- **3 x Dell PowerEdge R320 Intel Xeon E5-2430 v2 2.50GHz, 32GB RAM, 1TB HDD** {clu02, clu03, clu04} - 2015
- **1 x Dell PowerEdge R320 Intel Xeon E5-2430 v2 2.50GHz, 32GB RAM, 2 x 1TB HDD (RAID0)** {clu01} - **Master node** - 2015

Hortonworks Data Platform (HDP)

- SoftNet cluster runs the Hortonworks Data Platform (version 3.1.0)
- SoftNet cluster servers OS: Ubuntu Server 16.04 LTS
- HDP is a packaged software Hadoop distribution that aims to ease deployment and management of Hadoop clusters
- Compared with simply downloading the various Apache code bases and trying to run them together in a system, HDP greatly simplifies the use of Hadoop
- Architected, developed, and built completely in the open, HDP provides an enterprise ready data platform that enables organizations to adopt a Modern Data Architecture.
- With YARN as its architectural center it provides a data platform for multi-workload data processing across an array of processing methods
 - from batch through interactive to real-time, supported by key capabilities required of an enterprise data platform — spanning Governance, Security and Operations

HDP on SoftNet cluster managed by Ambari



HDP on SoftNet cluster managed by Ambari - Hosts

Ambari

Dashboard

Services

- HDFS
- YARN
- MapReduce2
- Tez
- Hive
- HBase
- Pig
- Sqoop
- ZooKeeper
- Storm
- Infra Solr
- Ambari Metrics
- Kafka
- Spark2
- Zeppelin Note...

Hosts

Alerts

Cluster Admin

Stack and Versions

Service Accounts

Kerberos

Service Auto Start

Hosts

softnetclu... 8 admin

Name	IP Address	Rack	Cores	RAM	Disk Usage	Load Avg	Versions	Components
clu01.softnet.tuc.gr	147.27.14.201	/default-rack	12 (12)	31.37GB		1.20	HDP-3.1.0.0	30 Components
clu02.softnet.tuc.gr	147.27.14.202	/default-rack	12 (12)	31.37GB		0.15	HDP-3.1.0.0	8 Components
clu03.softnet.tuc.gr	147.27.14.203	/default-rack	12 (12)	31.37GB		0.23	HDP-3.1.0.0	7 Components
clu04.softnet.tuc.gr	147.27.14.204	/default-rack	12 (12)	31.37GB		0.22	HDP-3.1.0.0	17 Components
clu05.softnet.tuc.gr	147.27.14.101	/default-rack	8 (8)	15.66GB		0.13	HDP-3.1.0.0	5 Components
clu06.softnet.tuc.gr	147.27.14.102	/default-rack	8 (8)	15.66GB		0.20	HDP-3.1.0.0	6 Components
clu07.softnet.tuc.gr	147.27.14.103	/default-rack	8 (8)	15.66GB			HDP-3.1.0.0	5 Components
clu08.softnet.tuc.gr	147.27.14.104	/default-rack	8 (8)	7.78GB		0.11	HDP-3.1.0.0	5 Components
clu09.softnet.tuc.gr	147.27.14.105	/default-rack	8 (8)	7.78GB		0.09	HDP-3.1.0.0	5 Components
clu10.softnet.tuc.gr	147.27.14.106	/default-rack	8 (8)	3.85GB			HDP-3.1.0.0	5 Components
clu11.softnet.tuc.gr	147.27.14.86	/default-rack	4 (4)	7.79GB		0.29	HDP-3.1.0.0	16 Components
clu12.softnet.tuc.gr	147.27.14.87	/default-rack	4 (4)	7.79GB		0.17	HDP-3.1.0.0	5 Components
clu13.softnet.tuc.gr	147.27.14.88	/default-rack	4 (4)	7.79GB		0.08	HDP-3.1.0.0	5 Components
clu14.softnet.tuc.gr	147.27.14.89	/default-rack	4 (4)	7.79GB			HDP-3.1.0.0	5 Components
clu15.softnet.tuc.gr	147.27.14.90	/default-rack	4 (4)	7.79GB		0.10	HDP-3.1.0.0	5 Components
clu16.softnet.tuc.gr	147.27.14.91	/default-rack	4 (4)	7.79GB		0.03	HDP-3.1.0.0	5 Components
clu17.softnet.tuc.gr	147.27.14.92	/default-rack	4 (4)	7.79GB		0.15	HDP-3.1.0.0	5 Components
clu19.softnet.tuc.gr	147.27.14.35	/default-rack	4 (4)	7.79GB		0.14	HDP-3.1.0.0	5 Components
clu20.softnet.tuc.gr	147.27.14.36	/default-rack	4 (4)	7.79GB		0.09	HDP-3.1.0.0	5 Components
clu24.softnet.tuc.gr	147.27.14.37	/default-rack	4 (4)	7.79GB		0.07	HDP-3.1.0.0	5 Components
clu25.softnet.tuc.gr	147.27.14.38	/default-rack	4 (4)	7.79GB		0.18	HDP-3.1.0.0	5 Components
clu26.softnet.tuc.gr	147.27.14.39	/default-rack	4 (4)	7.79GB		0.05	HDP-3.1.0.0	5 Components

Items per page: 50 1 - 22 of 22

Ambari

Dashboard

Services

Hosts

Alerts

Cluster Admin

Stack and Versions

Service Accounts

Kerberos

Service Auto Start

Admin / Stack

Stack

VERSIONS

Stack

Service	Version	Status	Description
HDFS	3.1.1	Installed	Apache Hadoop Distributed File System
YARN	3.1.1	Installed	Apache Hadoop NextGen MapReduce (YARN)
MapReduce2	3.1.1	Installed	Apache Hadoop NextGen MapReduce (YARN)
Tez	0.9.1	Installed	Tez is the next generation Hadoop Query Processing framework written on top of YARN
Hive	3.1.0	Installed	Data warehouse system for ad-hoc queries & analysis of large datasets and table & storage management service
HBase	2.0.2	Installed	Non-relational distributed database and centralized service for configuration management & synchronization
Pig	0.16.0	Installed	Scripting platform for analyzing large datasets
Sqoop	1.4.7	Installed	Tool for transferring bulk data between Apache Hadoop and structured data stores such as relational databases
Oozie	4.3.1	Add Service	System for workflow coordination and execution of Apache Hadoop jobs. This also includes the installation of the optional Oozie Web Console which relies on and will install the ExtJS Library.
ZooKeeper	3.4.6	Installed	Centralized service which provides highly reliable distributed coordination
Storm	1.2.1	Installed	Apache Hadoop Stream processing framework
Accumulo	1.7.0	Add Service	Robust, scalable, high performance distributed key/value store.
Infra Solr	0.1.0	Installed	Core shared service used by Ambari managed components.
Ambari Metrics	0.1.0	Installed	A system for metrics collection that provides storage and retrieval capability for metrics collected from the cluster
Atlas	1.1.0	Add Service	Atlas Metadata and Governance platform
Kafka	2.0.0	Installed	A high-throughput distributed messaging system
Knox	1.0.0	Add Service	Provides a single point of authentication and access for Apache Hadoop services in a cluster
Log Search	0.5.0	Add Service	Log aggregation, analysis, and visualization for Ambari managed services. This service is Technical Preview .
Ranger	1.2.0	Add Service	Comprehensive security for Hadoop
Ranger KMS	1.2.0	Add Service	Key Management Server
SmartSense	1.5.1.2.7.3.0-139	Add Service	SmartSense - Hortonworks SmartSense Tool (HST) helps quickly gather configuration, metrics, logs from common HDP services that aids to quickly troubleshoot support cases and receive cluster-specific recommendations.
Spark2	2.3.2	Installed	Apache Spark 2.3 is a fast and general engine for large-scale data processing.
Zeppelin Notebook	0.8.0	Installed	A web-based notebook that enables interactive data analytics. It enables you to make beautiful data-driven, interactive and collaborative documents with SQL, Scala and more.
Druid	0.12.1	Add Service	A fast column-oriented distributed data store.
Kerberos	1.10.3-30	Add Service	A computer network authentication protocol which works on the basis of 'tickets' to allow nodes communicating over a non-secure network to prove their identity to one another in a secure manner.
Superset	0.23.0	Add Service	Superset is a data exploration platform designed to be visual, intuitive and interactive. This service is Technical Preview .

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See third-party tools/resources that Ambari uses and their respective authors

Official Apache component versions for HDP 3.1.0

STACK VERSIONS[MANAGE VERSIONS](#)

FILTER: ALL (1)

	HDP-3.10.0 (3.1.0.0-78) Show Details
	CURRENT
HDFS	3.1.1
YARN	3.1.1
MapReduce2	3.1.1
Tez	0.9.1
Hive	3.1.0
HBase	2.0.2
Pig	0.16.0
Sqoop	1.4.7
ZooKeeper	3.4.6
Storm	1.2.1
Infra Solr	0.1.0
Ambari Metrics	0.1.0
Kafka	2.0.0
Spark2	2.3.2
Zeppelin Notebook	0.8.0

Apache
components
installed on
Softnet Cluster

YARN Resource Manager UI / Applications

←

↶

Not secure | clu01.softnet.tuc.gr:8189/ui2/#/yarn-apps/apps

🔍

🔗

☆

🌐

📄

📁

📱

🔧

🔒

👤

Cluster OverviewQueuesApplicationsServicesFlow ActivityNodesTools

Logged in as: UNKNOWN_USER

Home / Applications

Refresh

Re ▾Search...Search

12345Last - 4025 Rows ▾

User (13)

Filter

☒ tbitsakis589

☒ vvittis103

☒ livy101

☒ vkonidaris47

☒ akontaxakis46

☒ skatara40

More

State (3)

☒ FINISHED889

☒ KILLED101

☒ FAILED10

More

Apply

Clear

Application ID	Application Type	Application Name	User	State	Queue	Progress	Start Time	Elapsed Time
application_1651718580368_0001	TEZ	HIVE-42bc22b1...	hive	Finished	default	100%	2022/05/05 04:4...	10m 13s 822ms
application_1634491283298_0016	SPARK	Spark shell	xenia	Finished	default	100%	2022/04/12 08:2...	2h 40m 5s 761ms
application_1634491283298_0015	SPARK	Spark Pi	xenia	Finished	default	100%	2022/04/12 07:5...	16s 174ms
application_1634491283298_0014	SPARK	Spark shell	xenia	Finished	default	100%	2022/04/12 05:5...	4h 21m 51s 595...
application_1634491283298_0013	SPARK	Spark Pi	xenia	Finished	default	100%	2022/04/09 17:2...	22s 586ms
application_1634491283298_0012	SPARK	Spark Pi	xenia	Finished	default	100%	2022/04/09 10:4...	23s 118ms
application_1634491283298_0011	Apache Flink	Flink session clu...	akontaxakis	Finished	default	100%	2022/02/18 16:0...	2h 11m 25s 712ms
application_1634491283298_0010	Apache Flink	Flink session clu...	akontaxakis	Finished	default	100%	2022/02/15 11:4...	2h 11m 22s 804ms
application_1634491283298_0009	Apache Flink	Flink session clu...	akontaxakis	Finished	default	100%	2022/02/14 21:0...	2h 11m 24s 675ms
application_1634491283298_0008	Apache Flink	Flink session clu...	akontaxakis	Finished	default	100%	2022/02/14 13:2...	2h 11m 25s 870ms
application_1634491283298_0007	Apache Flink	Flink session clu...	akontaxakis	Finished	default	100%	2022/02/13 14:3...	2h 11m 32s 238ms
application_1634491283298_0006	Apache Flink	Flink session clu...	akontaxakis	Finished	default	100%	2022/02/12 12:0...	2h 11m 25s 979ms
application_1634491283298_0005	Apache Flink	Flink session clu...	akontaxakis	Finished	default	100%	2022/02/10 22:0...	2h 11m 26s 543ms
application_1634491283298_0004	Apache Flink	Flink session clu...	akontaxakis	Finished	default	100%	2022/02/10 19:1...	2h 11m 36s 348ms



Running jobs/
applications on
SoftNet cluster
(Hadoop MapReduce,
Spark, Flink)

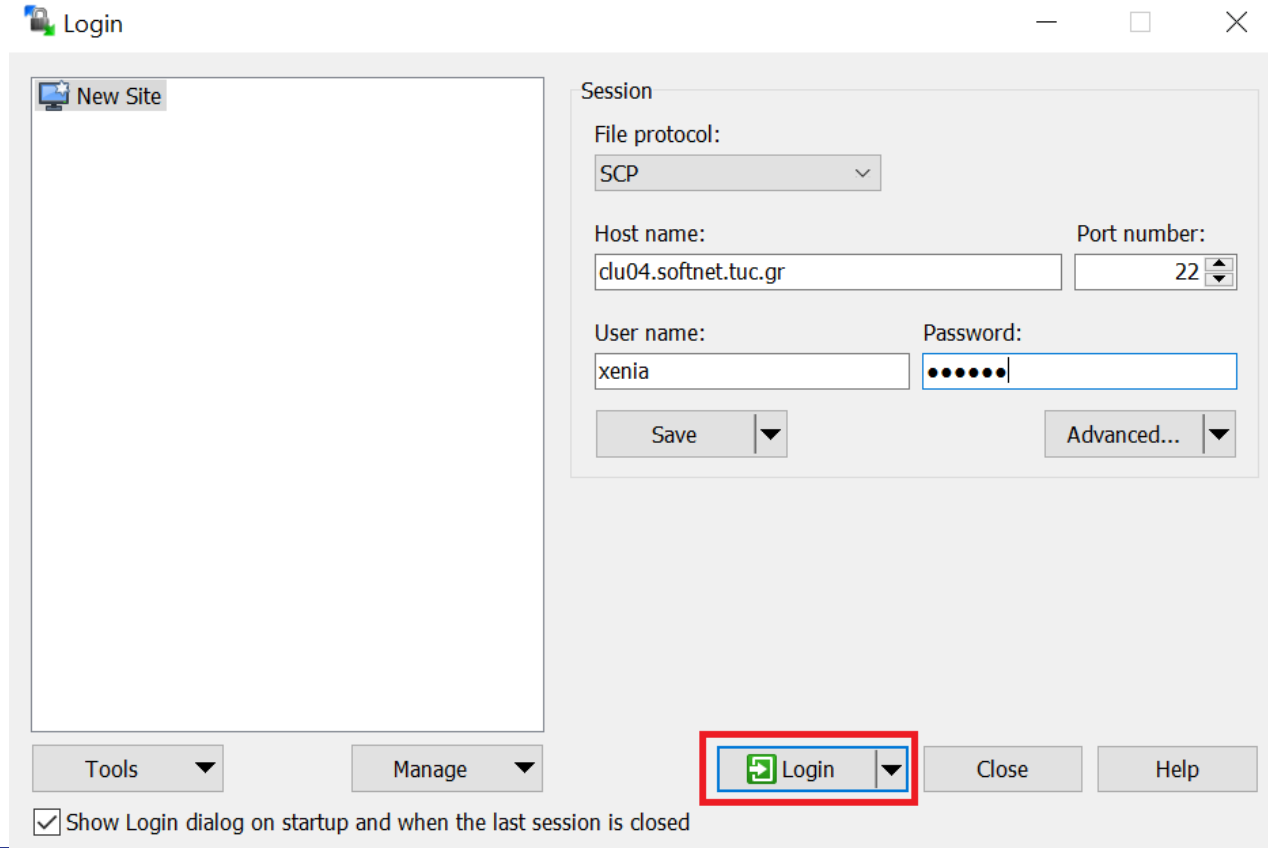
Upload files to SoftNet cluster / From Linux

- To submit jobs to the SoftNet cluster you must connect with your account* to `clu04.softnet.tuc.gr` via SSH (if you are outside TUC, you must connect to TUC VPN first)
- Open a terminal on your PC (Ctrl + Alt + T on Ubuntu)
- Upload a file (e.g. your Spark or Flink .jar) from your PC to your home directory on clu04:
`scp myjar.jar username@clu04.softnet.tuc.gr:/home/username/`
- Upload a folder from your PC to your home directory on clu04:
`scp -r myfolder username@clu04.softnet.tuc.gr:/home/username/`



Upload files to SoftNet cluster / From Windows

- If you are outside TUC, you must connect to TUC VPN first
- Install and use **WinSCP** to connect to clu04.softnet.tuc.gr and transfer the files from your PC to your home directory on clu04 (drag and drop)



Connect via SSH to SoftNet cluster / From Linux

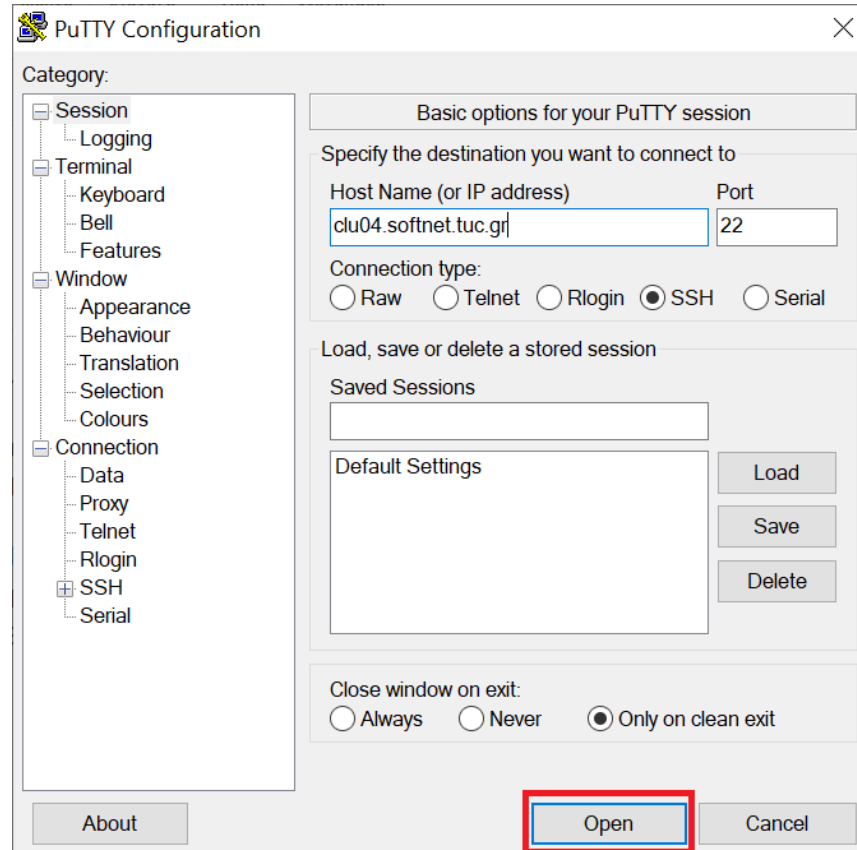
- If you are outside TUC, you must connect to TUC VPN first
- Open a terminal on your PC (Ctrl + Alt + T on Ubuntu)
- Connect to
clu04.softnet.tuc.gr via
SSH

ssh
username@clu04.softnet.tuc.gr

```
xenia@clu04: ~  
xenia@cleverdix:~$ ssh xenia@clu04.softnet.tuc.gr  
xenia@clu04.softnet.tuc.gr's password:  
Welcome to Ubuntu 16.04.6 LTS (GNU/Linux 4.4.0-210-generic x86_64)  
  
* Documentation:  https://help.ubuntu.com  
* Management:    https://landscape.canonical.com  
* Support:       https://ubuntu.com/advantage  
  
System information as of Thu May  5 16:12:48 EEST 2022  
  
System load:  0.19           Processes:            187  
Usage of /:   61.7% of 885.31GB Users logged in:     1  
Memory usage: 28%           IP address for eth0: 147.27.14.204  
Swap usage:   0%  
  
Graph this data and manage this system at:  
https://landscape.canonical.com/  
  
71 packages can be updated.  
4 updates are security updates.  
  
New release '18.04.6 LTS' available.  
Run 'do-release-upgrade' to upgrade to it.  
  
*** System restart required ***  
Last login: Thu May  5 16:12:28 2022 from 147.27.14.77  
xenia@clu04:~$
```

Connect via SSH to SoftNet cluster / From Windows

- To run jobs on SoftNet cluster you must connect with your account to clu04.softnet.tuc.gr via SSH (if you are outside TUC, you must connect to TUC VPN first)
- Install and use **Putty** on Windows to start an SSH session with clu04.softnet.tuc.gr



Submitting Spark jobs

Submitting Spark jobs

- To launch a Spark application in cluster mode:

```
cd /usr/hdp/current/spark2-client
```

```
./bin/spark-submit --class path.to.your.Class --master yarn --  
deploy-mode cluster [options] <app jar> [app options]
```

Note: There are two deploy modes that can be used to launch Spark applications on YARN:

- In **cluster mode**, the Spark driver runs inside an application master process which is managed by YARN on the cluster, and the client can go away after initiating the application.
- In **client mode**, the driver runs in the client process, and the application master is only used for requesting resources from YARN.

Unlike other cluster managers supported by Spark in which the master's address is specified in the `--master` parameter, in YARN mode the ResourceManager's address is picked up from the Hadoop configuration. Thus, the `--master` parameter is `yarn`.



Submitting Spark jobs

- For the needs of this tutorial we will use a given example .jar from HDP (folder /usr/hdp/current/spark2-client/examples/jars/):

```
cd /usr/hdp/current/spark2-client
```

```
./bin/spark-submit --class org.apache.spark.examples.SparkPi --  
master yarn --deploy-mode cluster --driver-memory 4g --num-  
executors 3 --executor-memory 2g --executor-cores 1  
examples/jars/spark-examples*.jar 10
```

SparkPi example calculates pi by “throwing darts” at a circle. The program generates points in the unit square ((0,0) to (1,1)) and counts how many points fall within the unit circle within the square. The result approximates pi.

Submitting Spark jobs

xenia@clu04: /usr/hdp/current/spark2-client

```
xenia@clu04:~$ cd /usr/hdp/current/spark2-client
```

```
xenia@clu04:/usr/hdp/current/spark2-client$ ./bin/spark-submit --class org.apache.spark.examples.SparkPi --master yarn --deploy-mode cluster --driver-memory 4g --num-executors 3 --executor-memory 2g --executor-cores 1 examples/jars/spark-examples*.jar 10
```

```
22/05/05 18:09:27 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
```

```
22/05/05 18:09:28 WARN DomainSocketFactory: The short-circuit local reads feature cannot be used because libhadoop cannot be loaded.
```

```
22/05/05 18:09:28 INFO RMPProxy: Connecting to ResourceManager at clu01.softnet.tuc.gr/147.27.14.201:8050
```

```
22/05/05 18:09:28 INFO Client: Requesting a new application from cluster with 18 NodeManagers
```

```
22/05/05 18:09:28 INFO Configuration: found resource resource-types.xml at file:/etc/hadoop/3.1.0.0-78/0/resource-types.xml
```

```
22/05/05 18:09:28 INFO Client: Verifying our application has not requested more than the maximum memory capability of the cluster (5120 MB per container)
```

```
22/05/05 18:09:28 INFO Client: Will allocate AM container, with 4505 MB memory including 409 MB overhead
```

```
22/05/05 18:09:28 INFO Client: Setting up container launch context for our AM
```

```
22/05/05 18:09:28 INFO Client: Setting up the launch environment for our AM container
```

```
22/05/05 18:09:28 INFO Client: Preparing resources for our AM container
```

```
22/05/05 18:09:29 INFO Client: Use hdfs cache file as spark.yarn.archive for HDP, hdfsCacheFile:hdfs://clu01.softnet.tuc.gr:8020/hdp/apps/3.1.0.0-78/spark2/spark2-hdp-yarn-archive.tar.gz
```

```
22/05/05 18:09:29 INFO Client: Source and destination file systems are the same. Not copying hdfs://clu01.softnet.tuc.gr:8020/hdp/apps/3.1.0.0-78/spark2/spark2-hdp-yarn-archive.tar.gz
```

```
22/05/05 18:09:29 INFO Client: Distribute hdfs cache file as spark.sql.hive.metastore.jars for HDP, hdfsCacheFile:hdfs://clu01.softnet.tuc.gr:8020/hdp/apps/3.1.0.0-78/spark2/spark2-hdp-hive-archive.tar.gz
```

```
22/05/05 18:09:29 INFO Client: Source and destination file systems are the same. Not copying hdfs://clu01.softnet.tuc.gr:8020/hdp/apps/3.1.0.0-78/spark2/spark2-hdp-hive-archive.tar.gz
```

```
22/05/05 18:09:29 INFO Client: Uploading resource file:/usr/hdp/3.1.0.0-78/spark2/examples/jars/spark-examples_2.11-2.3.2.3.1.0.0-78.jar -> hdfs://clu01.softnet.tuc.gr:8020/user/xenia/.sparkStaging/application_1651718580368_006/spark-examples_2.11-2.3.2.3.1.0.0-78.jar
```

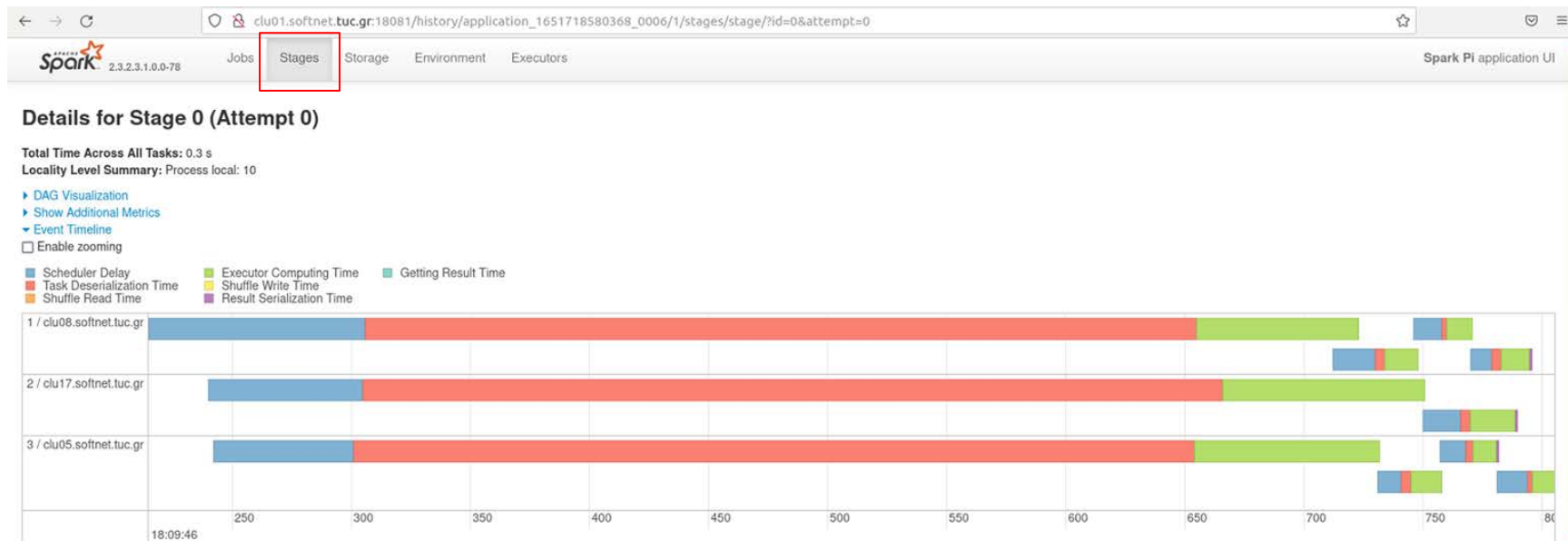
```
22/05/05 18:09:30 INFO Client: Uploading resource file:/tmp/spark-9daf148d-b932-48af-8f98-2fb7616406db/___spark_conf_1419837843114352582.zip -> hdfs://clu01.softnet.tuc.gr:8020/user/xenia/.sparkStaging/application_1651718580368_0006/___spark_conf_1419837843114352582.zip
```

```
22/05/05 18:09:30 INFO Configuration: Checking user roles for xenia
```

Submitting Spark jobs

```
xenia@clu04: /usr/hdp/current/spark2-client
22/05/05 18:09:39 INFO Client: Application report for application_1651718580368_0006 (state: RUNNING)
22/05/05 18:09:39 INFO Client:
  client token: N/A
  diagnostics: N/A
  ApplicationMaster host: 147.27.14.39
  ApplicationMaster RPC port: 0
  queue: default
  start time: 1651763370549
  final status: UNDEFINED
  tracking URL: http://clu01.softnet.tuc.gr:8189/proxy/application_1651718580368_0006/
  user: xenia
22/05/05 18:09:40 INFO Client: Application report for application_1651718580368_0006 (state: RUNNING)
22/05/05 18:09:41 INFO Client: Application report for application_1651718580368_0006 (state: RUNNING)
22/05/05 18:09:42 INFO Client: Application report for application_1651718580368_0006 (state: RUNNING)
22/05/05 18:09:43 INFO Client: Application report for application_1651718580368_0006 (state: RUNNING)
22/05/05 18:09:44 INFO Client: Application report for application_1651718580368_0006 (state: RUNNING)
22/05/05 18:09:45 INFO Client: Application report for application_1651718580368_0006 (state: RUNNING)
22/05/05 18:09:46 INFO Client: Application report for application_1651718580368_0006 (state: RUNNING)
22/05/05 18:09:47 INFO Client: Application report for application_1651718580368_0006 (state: FINISHED)
22/05/05 18:09:47 INFO Client:
  client token: N/A
  diagnostics: N/A
  ApplicationMaster host: 147.27.14.39
  ApplicationMaster RPC port: 0
  queue: default
  start time: 1651763370549
  final status: SUCCEEDED
  tracking URL: http://clu01.softnet.tuc.gr:8189/proxy/application_1651718580368_0006/
  user: xenia
22/05/05 18:09:47 INFO ShutdownHookManager: Shutdown hook called
22/05/05 18:09:47 INFO ShutdownHookManager: Deleting directory /tmp/spark-9daf148d-b932-48af-8f98-2fb7616406db
22/05/05 18:09:47 INFO ShutdownHookManager: Deleting directory /tmp/spark-3217e9e0-41c0-459b-a241-8522dff0207
xenia@clu04: /usr/hdp/current/spark2-client$
```

Submitting Spark jobs



Summary Metrics for 10 Completed Tasks

Metric	Min	25th percentile	Median	75th percentile	Max
Duration	10 ms	11 ms	14 ms	68 ms	85 ms
GC Time	0 ms	0 ms	0 ms	0 ms	9 ms

▼ Aggregated Metrics by Executor

Executor ID ▲	Address	Task Time	Total Tasks	Failed Tasks	Killed Tasks	Succeeded Tasks	Blacklisted
1	stdout stderr clu08.softnet.tuc.gr:46311	0.6 s	4	0	0	4	false
2	stdout stderr clu17.softnet.tuc.gr:43833	0.6 s	2	0	0	2	false
3	stdout stderr clu05.softnet.tuc.gr:32810	0.6 s	4	0	0	4	false

Submitting Spark jobs/ Yarn Resource Manager UI

← → ↺ 🔒 Not secure | clu01.softnet.tuc.gr:8189/ui2/#/yarn-apps/apps 🔍 ↗ ☆ 🌐 📄 📁 📌 📧 ⚙️ 🗑️ 👤

 [Cluster Overview](#) [Queues](#) [Applications](#) [Services](#) [Flow Activity](#) [Nodes](#) [Tools](#) Logged in as: UNKNOWN_USER

[Home](#) / [Applications](#) Refresh

Re ▾ Search... Search 1 2 3 4 5 Last - 40 25 Rows ▾

User (13)	Application ID	Application Type	Application Name	User	State	Queue	Progress	Start Time	Elapsed Time
Filter	application_1651718580368_0006	SPARK	org.apache.spar...	xenia	Finished	default	100%	2022/05/05 17:0...	16s 882ms
☒ tbitsakis 584	application_1651718580368_0005	SPARK	org.apache.spar...	xenia	Finished	default	100%	2022/05/05 07:1...	27s 243ms
☒ vvittis 103	application_1651718580368_0004	SPARK	org.apache.spar...	xenia	Failed	thequeue	0%	2022/05/05 07:1...	39ms
☒ vkonidaris 47	application_1651718580368_0003	SPARK	org.apache.spar...	xenia	Failed	thequeue	0%	2022/05/05 07:1...	32ms
☒ akontaxakis 46	application_1651718580368_0002	SPARK	Spark Pi	xenia	Finished	default	100%	2022/05/05 07:0...	21s 544ms
☒ skatara 40 More	application_1651718580368_0001	TEZ	HIVE-42bc22b1-...	hive	Finished	default	100%	2022/05/05 04:4...	10m 13s 822ms
State (3)	application_1634491283298_0016	SPARK	Spark shell	xenia	Finished	default	100%	2022/04/12 08:2...	2h 40m 5s 761ms
☒ FINISHED 887	application_1634491283298_0015	SPARK	Spark Pi	xenia	Finished	default	100%	2022/04/12 07:5...	16s 174ms
☒ KILLED 101	application_1634491283298_0014	SPARK	Spark shell	xenia	Finished	default	100%	2022/04/12 05:5...	4h 21m 51s 595...
☒ FAILED 12 Less	application_1634491283298_0013	SPARK	Spark Pi	xenia	Finished	default	100%	2022/04/09 17:2...	22s 586ms
Queue (2)	application_1634491283298_0012	SPARK	Spark Pi	xenia	Finished	default	100%	2022/04/09 10:4...	23s 118ms
☒ default 998	application_1634491283298_0011	Apache Flink	Flink session clu...	akontaxakis	Finished	default	100%	2022/02/18 16:0...	2h 11m 25s 712ms
☒ thequeue 2 More	application_1634491283298_0010	Apache Flink	Flink session clu...	akontaxakis	Finished	default	100%	2022/02/15 11:4...	2h 11m 22s 804ms

Apply Clear

Submitting Spark jobs/ Yarn Resource Manager UI

The screenshot shows the Databricks web interface. At the top, there's a navigation bar with links like 'Cluster Overview', 'Queues', 'Applications', 'Services', 'Flow Activity', 'Nodes', and 'Tools'. The user is logged in as 'UNKNOWN_USER'. The main content area displays the application 'org.apache.spark.examples.SparkPi' with a status of 'SUCCEEDED'. Below this, there's a section for 'Application Attempts' with tabs for 'Attempts List', 'Resource Usage', 'Diagnostics', and 'Logs'. The 'Attempts List' tab is active, showing a timeline of attempts. A red arrow points to the 'History' link in the top right corner of the application details section.

Submitting Spark jobs (client deploy mode)

WordCount
example

```
xenia@clu04: /usr/hdp/current/spark2-client
xenia@clu04:/usr/hdp/current/spark2-client$ hadoop fs -copyFromLocal /etc/hadoop/conf/log4j.properties /tmp/data
xenia@clu04:/usr/hdp/current/spark2-client$ ./bin/spark-shell --master yarn --deploy-mode client --driver-memory 512m --executor-memory 512m
Setting default log level to "WARN".
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLogLevel(newLevel).
Spark context Web UI available at http://clu04.softnet.tuc.gr:4040
Spark context available as 'sc' (master = yarn, app id = application_1651718580368_0007).
Spark session available as 'spark'.
Welcome to

      /_/_/_/_/_/_/_/_/_/_/_/_/_/_/_/_\
     /_/_/_/_/_/_/_/_/_/_/_/_/_/_/_/_\
    /_/_/_/_/_/_/_/_/_/_/_/_/_/_/_/_\
   /_/_/_/_/_/_/_/_/_/_/_/_/_/_/_/_\
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 /_/_/_/_/_/_/_/_/_/_/_/_/_/_/_/_\
/_/_/_/_/_/_/_/_/_/_/_/_/_/_/__\

version 2.3.2.3.1.0.0-78

ID *      Address      Task Time      Total Tasks      Failed Tasks
Using Scala version 2.11.8 (Java HotSpot(TM) 64-Bit Server VM, Java 1.8.0_112)
Type in expressions to have them evaluated.
Type :help for more information.
scala> val file = sc.textFile("/tmp/data")
file: org.apache.spark.rdd.RDD[String] = /tmp/data MapPartitionsRDD[1] at textFile at <console>:24

scala> val counts = file.flatMap(line => line.split(" ")).map(word => (word, 1)).reduceByKey(_ + _)
counts: org.apache.spark.rdd.RDD[(String, Int)] = ShuffledRDD[4] at reduceByKey at <console>:25

scala> counts.saveAsTextFile("/tmp/wordcount")

ID *      Attempt *      Status      Locality Level      Executor ID      Host
0 *      0 *      SUCCESS      PROCESS_LOCAL      1 *      clu08.softnet.tuc.gr

scala> counts.count()
res1: Long = 364
```


Submitting Hadoop MapReduce jobs

Submitting Hadoop MapReduce jobs via Yarn

```
xenia@clu04:~$ cd /usr/hdp/current/hadoop-mapreduce-client
```

```
xenia@clu04:/usr/hdp/current/hadoop-mapreduce-client$ yarn jar  
hadoop-mapreduce-examples.jar
```

```
xenia@clu04:/usr/hdp/current/hadoop-mapreduce-client$ yarn jar  
hadoop-mapreduce-examples.jar pi 16 100000
```



Submitting Flink jobs

Flink on SoftNet cluster

- Flink is not integrated into HDP and has been installed on SoftNet cluster separately / manually
- Flink 1.10.0 (for Scala 2.12)
- Flink installation path: /usr/local/flink
- 1 Flink cluster master: clu01.softnet.tuc.gr
- 11 Flink cluster slaves with similar system characteristics:
{clu26, clu25, clu24, clu20, clu19, clu17, clu16, clu15, clu13, clu12, clu11} .softnet.tuc.gr
- Flink Web Front End: <http://clu01.softnet.tuc.gr:8081>



Flink on SoftNet cluster

- Flink Web Front End: <http://clu01.softnet.tuc.gr:8081>

The screenshot shows the Apache Flink Dashboard Overview page. The left sidebar contains navigation links: Overview (selected), Jobs, Running Jobs, Completed Jobs, Task Managers, Job Manager, and Submit New Job. The main content area displays the following information:

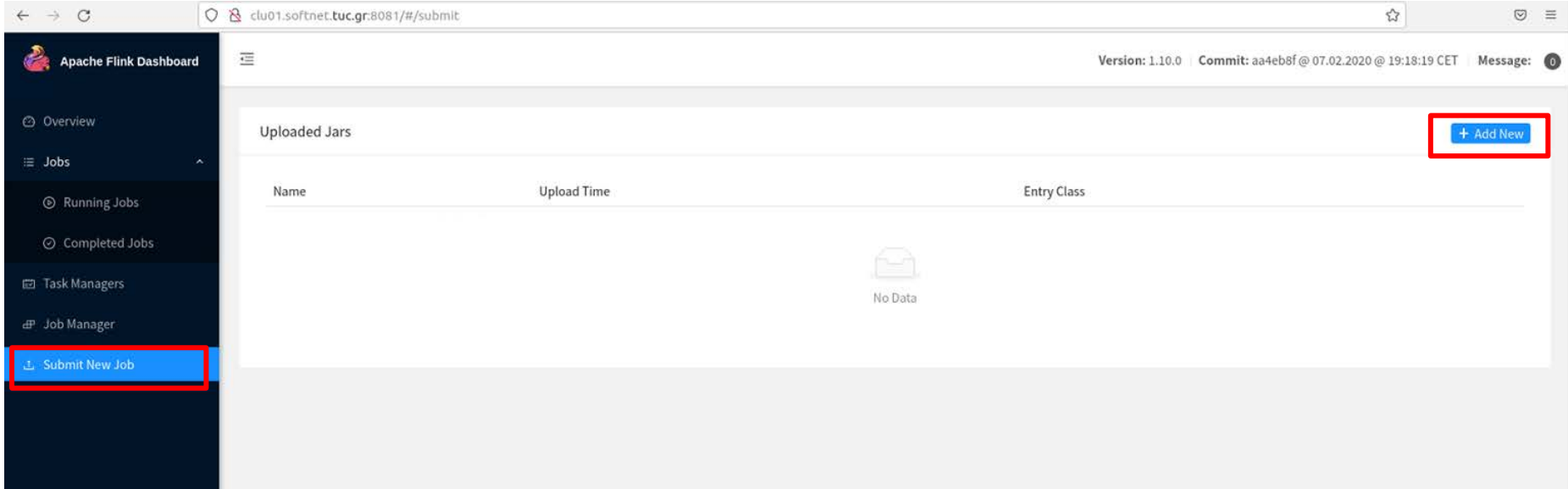
- Available Task Slots:** 33 (Total Task Slots: 33, Task Managers: 11)
- Running Jobs:** 0 (Finished: 0, Canceled: 0, Failed: 0)
- Running Job List:** A table with columns: Job Name, Start Time, Duration, End Time, Tasks, and Status. The table is empty, showing "No Data".
- Completed Job List:** A table with columns: Job Name, Start Time, Duration, End Time, Tasks, and Status. The table is empty.

Submitting Flink jobs

- Submitting a job means uploading the job's JAR and related dependencies to the Flink cluster and initiating the job execution
 - For the sake of this tutorial, we select an example .jar from Apache Flink available at /usr/local/flink/examples/streaming/, the WordCount.jar
 - WordCount computes the frequency of words in a text collection. The algorithm works in two steps: First, splits the text into individual words. Second, the words are grouped and counted.
 - More info on these examples can be found at: <https://nightlies.apache.org/flink/flink-docs-master/docs/dev/dataset/examples/>
 - Feel free to select any other JAR archive from the examples/ folder or deploy your own job, as you are going to do with your project
- 

Submitting Flink jobs (1st method - Dashboard)

- Flink Web Front End: <http://clu01.softnet.tuc.gr:8081>

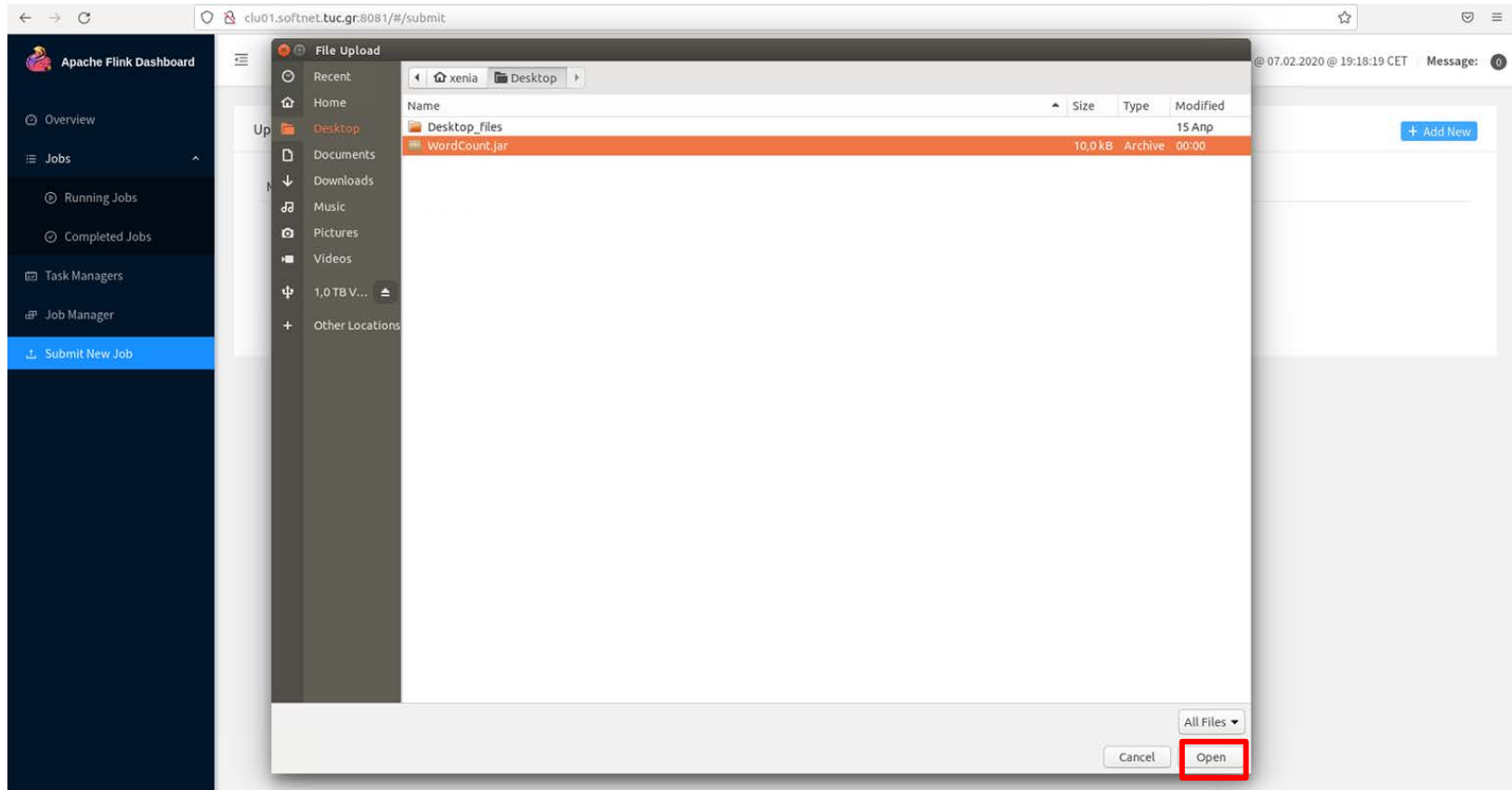


The screenshot displays the Apache Flink Dashboard interface. The browser address bar shows the URL `clu01.softnet.tuc.gr:8081/#/submit`. The dashboard header includes the Apache Flink logo and the text "Apache Flink Dashboard". The top right corner displays the version "Version: 1.10.0", the commit "Commit: aa4eb8f @ 07.02.2020 @ 19:18:19 CET", and a message count "Message: 0".

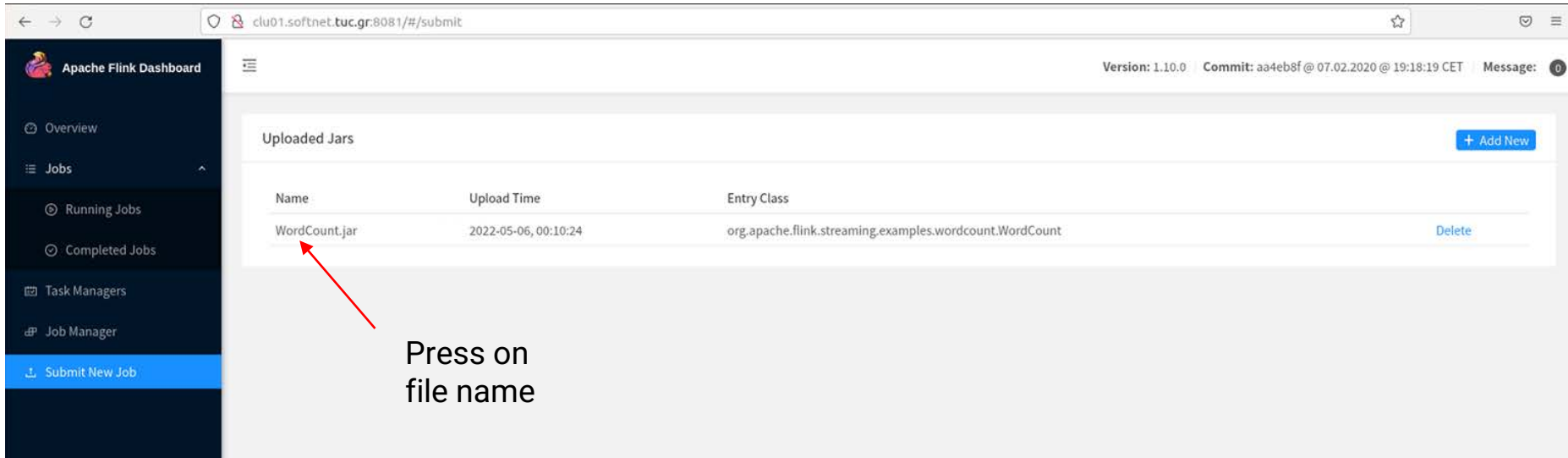
The left sidebar contains a navigation menu with the following items: Overview, Jobs, Running Jobs, Completed Jobs, Task Managers, Job Manager, and Submit New Job. The "Submit New Job" item is highlighted with a red box.

The main content area is titled "Uploaded Jars" and features a table with columns for Name, Upload Time, and Entry Class. The table is currently empty, displaying a "No Data" message. A red box highlights the "+ Add New" button in the top right corner of the "Uploaded Jars" section.

Submitting Flink jobs (1st method - Dashboard)



Submitting Flink jobs (1st method - Dashboard)



The screenshot shows the Apache Flink Dashboard interface. The left sidebar contains navigation links: Overview, Jobs (expanded), Running Jobs, Completed Jobs, Task Managers, Job Manager, and Submit New Job (highlighted in blue). The main content area displays the 'Uploaded Jars' section, which includes a table with the following data:

Name	Upload Time	Entry Class	
WordCount.jar	2022-05-06, 00:10:24	org.apache.flink.streaming.examples.wordcount.WordCount	Delete

A red arrow points to the 'WordCount.jar' file name in the table, and the text 'Press on file name' is displayed next to it. The top right of the dashboard shows the version (1.10.0), commit (aa4eb8f), and timestamp (07.02.2020 @ 19:18:19 CET).

Submitting Flink jobs (1st method - Dashboard)

← → ↻ Not secure | clu01.softnet.tuc.gr:8081/#/submit

Apache Flink Dashboard

Version: 1.10.0 Commit: aa4eb8f @ 07.02.2020 @ 19:18:19 CET Message: 1

Overview

Jobs

- Running Jobs
- Completed Jobs

Task Managers

Job Manager

Submit New Job

Uploaded Jars

+ Add New

Name	Upload Time	Entry Class
WordCount.jar	2022-05-06, 00:10:24	org.apache.flink.streaming.examples.wordcount.WordCount

org.apache.flink.streaming.examples.wordcount.WordCount 3

Program Arguments

Savepoint Path

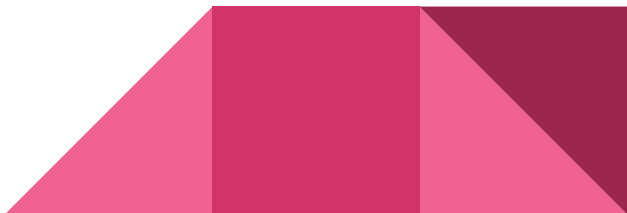
☐ Allow Non Restored State

Show Plan **Submit**

 Apache Flink Dashboard

- Overview
- Jobs
- Running Jobs
- **Completed Jobs**
- Task Managers
- Job Manager
- Submit New Job

Version: 1.10.0 Commit: aa4eb8f @ 07.02.2020 @ 19:18:19 CET Message: 1



Submitting Flink jobs (1st method - Dashboard)

← → ↺ ⚠ Not secure | clu01.softnet.tuc.gr:8081/#/job/8ab174342e719821374751d7bab1e4aa/overview 🔍 📄 ⬆ 📺 ⚙️ 🗑️ 👤 ⋮

Apache Flink Dashboard ☰ Version: 1.10.0 Commit: aa4eb8f @ 07.02.2020 @ 19:18:19 CET Message: 1

[Overview](#) [Jobs](#) [Running Jobs](#) [Completed Jobs](#) [Task Managers](#) [Job Manager](#) [Submit New Job](#)

Streaming WordCount FINISHED 3

ID: 8ab174342e719821374751d7bab1e4aa | Start Time: 2022-05-06 01:18:33 | End Time: 2022-05-06 01:18:34 | Duration: 791ms

[Overview](#) [Exceptions](#) [TimeLine](#) [Checkpoints](#) [Configuration](#)

```
graph LR; A["Source: Collection Source  
Parallelism: 1"] -- REBALANCE --> B["Flat Map  
Parallelism: 3"]; B -- HASH --> C["Keyed Aggregation -> Sink: Print to Std. Out  
Parallelism: 3"]
```

Name	Status	Bytes Received	Records Received	Bytes Sent	Records Sent	Parallelism	Start Time	Tasks
Source: Collection Source	FINISHED	0 B	0	0 B	35	1	2022-05-06 01:18	1
Flat Map	FINISHED	1.66 KB	35	0 B	287	3	2022-05-06 01:18	3
Keyed Aggregation -> Sink: Print to Std. Out	FINISHED	3.97 KB	287	0 B	0	3	2022-05-06 01:18	3

Submit Flink jobs (2nd method - Command-line interface)

- Upload the Flink project .jar from your PC to your home directory on clu04.softnet.tuc.gr, as already shown on previous slides.

- Connect to clu04.softnet.tuc.gr via SSH

- Submit the Flink job:

```
cd /usr/local/flink/
```

```
./bin/flink run /usr/local/flink/examples/batch/WordCount.jar
```

- The other examples can be started in a similar way.
- Note that many examples run without passing any arguments for them, by using build-in data. To run WordCount with real data, you have to pass the path to the data:

```
./bin/flink run ./examples/batch/WordCount.jar --input  
/path/to/some/text/data --output /path/to/result
```

- Non-local file systems require a schema prefix, such as hdfs://.

Submit Flink jobs (2nd method - Command-line interface)

```
softnet@clu01: /usr/local/flink

softnet@clu01:~$ cd /usr/local/flink/
softnet@clu01:/usr/local/flink$ ./bin/flink run /usr/local/flink/examples/batch/WordCount.jar
SLF4J: Class path contains multiple SLF4J bindings.
SLF4J: Found binding in [jar:file:/usr/local/flink-1.10.0/lib/slf4j-log4j12-1.7.15.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: Found binding in [jar:file:/usr/hdp/3.1.0.0-78/hadoop/lib/slf4j-log4j12-1.7.25.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: See http://www.slf4j.org/codes.html#multiple_bindings for an explanation.
SLF4J: Actual binding is of type [org.slf4j.impl.Log4jLoggerFactory]
Executing WordCount example with default input data set.
Use --input to specify file input.
Printing result to stdout. Use --output to specify output path.
Job has been submitted with JobID 708d1305e92e5ca96f00c78d386026cb
Program execution finished
Job with JobID 708d1305e92e5ca96f00c78d386026cb has finished.
Job Runtime: 904 ms
Accumulator Results:
- b7ec94f03f415322fc24a99aa0ebbf84 (java.util.ArrayList) [170 elements]

(a,5)
(action,1)
(after,1)
(against,1)
(all,2)
(and,12)
(arms,1)
(arrows,1)
(awry,1)
(ay,1)
(bare,1)
(be,4)
(bear,3)
```

Replace
softnet@clu01
with
yourusername
@clu04

Kafka

SoftNet cluster Kafka brokers and Zookeeper servers

- 4 Kafka brokers: {clu02, clu03, clu04, clu06}.softnet.tuc.gr, Port: 6667
- 3 Zookeeper servers: {clu01, clu02, clu03}.softnet.tuc.gr, Port: 2182



Create & list Kafka topics

- Create a kafka topic:

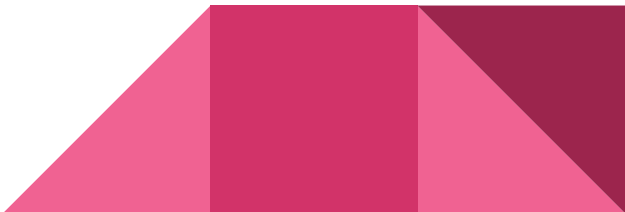
```
xenia@clu04:~$ cd /usr/hdp/current/kafka-broker
```

```
xenia@clu04:/usr/hdp/current/kafka-broker$ bin/kafka-topics.sh -  
-create --zookeeper clu01.softnet.tuc.gr:2182 --replication-  
factor 2 --partitions 2 --topic kafkatest
```

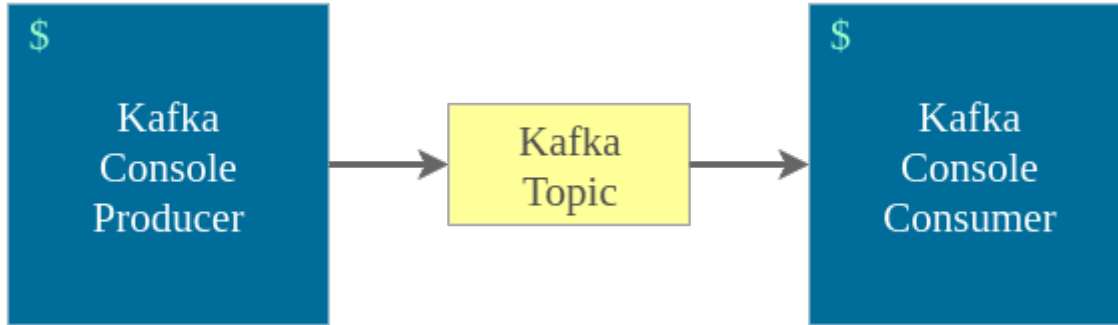
```
Created topic "kafkatest".
```

- List kafka topics:

```
xenia@clu04:/usr/hdp/current/kafka-broker$ bin/kafka-topics.sh -  
-list --zookeeper clu01.softnet.tuc.gr:2182
```



Kafka producer/consumer example



Kafka producer/consumer example

- **Run the producer script.** Once it starts, we can enter lines to standard input and they will be published to Kafka:

```
xenia@clu04:/usr/hdp/current/kafka-broker$ bin/kafka-  
console-producer.sh --broker-list  
clu02.softnet.tuc.gr:6667,clu03.softnet.tuc.gr:6667,clu04  
.softnet.tuc.gr:6667,clu06.softnet.tuc.gr:6667 --topic  
kafkatest
```

>

**** Note** that unlike the other Kafka scripts which specify a Zookeeper instance, the publisher is specifying a broker list.

Kafka producer/consumer example

- **Run the sample consumer script.** If things go well, you should see the messages that you typed in the producer window show up on the consumer terminal.
- To run the consumer script, open another terminal on your PC and ssh to clu04.softnet.tuc.gr. Then run:

```
xenia@clu04:~$ cd /usr/hdp/current/kafka-broker
```

```
xenia@clu04:/usr/hdp/current/kafka-broker$ bin/kafka-console-consumer.sh  
--bootstrap-server
```

```
clu02.softnet.tuc.gr:6667,clu03.softnet.tuc.gr:6667,clu04.softnet.tuc.gr:  
6667,clu06.softnet.tuc.gr:6667 --topic kafkatest --from-beginning
```



Kafka producer/consumer example

```
xenia@clu04: /usr/hdp/current/kafka-broker
xenia@clu04:/usr/hdp/current/kafka-broker$ bin/kafka-topics.sh --create --zookeeper clu01.softnet.tuc.gr:2182 --r
eplication-factor 2 --partitions 2 --topic kafkatest
Created topic "kafkatest".
xenia@clu04:/usr/hdp/current/kafka-broker$ bin/kafka-console-producer.sh --broker-list clu02.softnet.tuc.gr:6667,
clu03.softnet.tuc.gr:6667,clu04.softnet.tuc.gr:6667,clu06.softnet.tuc.gr:6667 --topic kafkatest
>I
>Love
>SoftNet
>Cluster
>.
>[]

xenia@clu04: /usr/hdp/current/kafka-broker
xenia@clu04:/usr/hdp/current/kafka-broker$ bin/kafka-console-consumer.sh --bootstrap-server clu02.softnet.tuc.gr:
6667,clu03.softnet.tuc.gr:6667,clu04.softnet.tuc.gr:6667,clu06.softnet.tuc.gr:6667 --topic kafkatest --from-begin
ning
Love
Cluster
I
SoftNet
.
```

Delete Kafka topics

```
xenia@clu04:/usr/hdp/current/kafka-broker$ bin/kafka-topics.sh -  
-delete --zookeeper clu01.softnet.tuc.gr:2182 --topic kafkatest
```

Topic kafkatest is marked for deletion.

Note: This will have no impact if `delete.topic.enable` is not set to `true`.

